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Analytics And Data Science

To Recognize Risks Earlier, Invest in Analytics

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You've probably heard business leaders justify their flat-footedness in a crisis by claiming that every organization is flying blind in times of deep uncertainty. But in fact some leaders know precisely where they're going. They understand what's required to chart a course through market turbulence, and they've built organizations with keen situational awareness.

When it comes to developing the ability to figure out where things are heading and respond nimbly to a changing environment, nothing is more important than analytics. Unfortunately, in recent years analytics (also known as data mining or business intelligence) has become the unloved stepchild of data sciences, overshadowed by machine learning and statistics. Those two disciplines layer mathematical sophistication on top of a foundation of human intuition, creating an appealing illusion of objectivity and deft steering. Ironically, of the three, analytics is the most essential competency for navigating crises.

Solutions based on AI and machine learning hum along well during stable times but fall apart when disaster strikes. These technologies automate tasks by extracting patterns from data and turning them into instructions. Such models can quickly become obsolete when the inputs to the system change. Analytics, in contrast, alerts you when the rules of the game are changing. Without that kind of a warning, automation solutions can quickly go off the rails, leaving you exposed to exogenous shocks.

Statistics has a similar shortcoming during a crisis. Statisticians help decision-makers get rigorous answers. But what if they're asking the wrong questions? While statistical skills are required to

test hypotheses, analysts have the acumen to come up with the right hypotheses in the first place. To attempt statistics without analytics, you'd need great confidence in your assumptions—the kind of confidence that's foolhardy when a crisis pulls the rug out from under you.

Analysts thrive in ambiguity. Their talent is exploration, which makes them particularly good at foreseeing and responding to crises. By searching internal and external data sources for critical information, analysts keep a finger on the pulse of what's going on. They scan the horizon for trends and formulate questions about what's behind them. Their job is to inspire executives with thought-provoking yet qualified possibilities. Once the highest-priority hypotheses have been short-listed by leaders, then it's time to call in a statistician to pressure-test them and separate true insights from red herrings.

During good times, leading organizations build analytics capabilities to strengthen their ability to innovate. Analysts' ability to find clues to such things as shifting consumer tastes can help firms take advantage of opportunities before less-savvy competitors do. When the going gets tough, however, what looked like a nice-to-have innovation booster turns into a must-have safety net. To be sure, some events are impossible to see in advance—the true black swans—but addressing their fallout is a game best played with open eyes.

Unfortunately, it's very hard to cobble together a mature analytics department on short notice. The technical skills that allow analysts to guzzle data with lightning speed merely increase the mass of information they encounter. Spotting a gem in it takes something more. Without domain knowledge, business acumen,

and strong intuition about the practical value of discoveries—as well as the communication skills to convey them to decision-makers effectively—analysts will struggle to be useful. It takes time for them to learn to judge what’s important in addition to what’s interesting. You can’t expect them to be an instant solution to charting a course through your latest crisis. Instead, see them as an investment in your future nimbleness.

It also takes time to secure access to the promising data sources analysts need. Ideally, business leaders won’t wait for a big disruption to begin building relationships with data vendors, industry partners, and data collection specialists. Bear in mind that in the face of an extreme shock, your historical data sources may become obsolete. If your understanding of the past fails to give you a useful window on tomorrow’s world—perhaps because a pandemic has changed everything—it doesn’t matter how good your information was yesterday. You need new information. After the 2008 financial crash, for example, banks around the world recognized that there might be an advantage to analyzing nontraditional signals of creditworthiness, such as data from supermarket loyalty cards, but not all players were equally positioned to get access to them.



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Additionally, your internal data stores may require special processing before analysts can mine them, so it's worth thinking about hiring supporting data engineers. If analytics is the discipline of making data useful, then data engineering is the

discipline of making data usable; it provides behind-the-scenes infrastructure that makes machine logs and colossal data stores compatible with analytics tool kits.

When I began speaking at conferences about the importance of analytics, I found that convincing an audience of its value was the easy part. The mood changed when I explained the catch: Analytics is a time investment. You can't count on getting something useful out of every foray into a data set. To succeed at exploration, your organization needs a culture of no-strings-attached analytics. As the leader, you are responsible for setting the scope (which data sources should be looked at) and the time frame ("You have two weeks to explore this database"). Then you must ensure that analysts aren't punished for coming back empty-handed.

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Once business leaders accept that analytics represents an investment that may not immediately pay off, I hit the next stumbling block: the perception that only a large and technologically sophisticated company such as Alphabet can afford it. This is nonsense. In my experience you're more likely to find analytics thriving in start-ups than at well-established behemoths.

Start-ups naturally invest in analytics as they try to navigate a new market, with several generalists taking on a share of the exploratory work. Then as the venture grows, the culture changes. Workers are trusted less and made more accountable for return on their efforts, and overzealous management stifles opportunities for analytics to thrive. Analysts hired into this culture rarely get to enjoy the most interesting part of their work—exploration—and instead serve as human search engines and dashboard janitors. Many quit out of frustration as their potential is squandered.

Creating a culture where analytics flourishes takes thoughtful leadership. As organizations grow toward incumbency, only the most visionary will have the courage to nurture a true analytics department and make sure that business leaders have access to it and are influenced by it. Industries that have been burned by a previous crisis—banking is a good example—are especially likely to invest in analytics and apply it to risk management.

Becoming a leader in analytics takes a commitment to trust your analysts and give them space to do their work. Their job, after all, will be to reveal threats that you never even imagined should be on your radar. That sort of work can't be managed with a stopwatch and a checklist.

Crises such as a pandemic—when no one has the answers, and uncertainty is high—remind us of the importance of asking the right questions. Analytics gives firms an edge in learning and adapting. When the world is suddenly upended, those who can learn the fastest are best positioned to succeed. Smart companies will invest in analytics today to get ahead of whatever is coming tomorrow.

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